

QUESTION 1 (9 MARKS)

	Marks
(a) (i) If $2 \tan^2 x - \sec^2 x = 2$, find the exact values of $\tan x$	2
(ii) Hence, find the values of x in the domain $0 \leq x \leq 2\pi$	2
(b) The gradient of a function is given by $\frac{dy}{dx} = \sqrt{x} - \frac{1}{\sqrt{x}}$ and it passes through the point (4,3). Find the equation of the function.	3
(c) Solve for x : $\frac{x}{x+3} > 1$	2

QUESTION 2 (9 Marks) – START A NEW PAGE

	Marks
(a) Using the substitution $u^2 = x$, where $u > 0$, evaluate $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$	3
(b) If $\sin^{-1} t$, $\cos^{-1} t$, and $\sin^{-1}(1 - t)$ are acute	
(i) Show that $\sin(\sin^{-1} t - \cos^{-1} t) = 2t^2 - 1$ for $0 < t < 1$.	2
(ii) Hence, solve the equation: $\sin^{-1} t - \cos^{-1} t = \sin^{-1}(1 - t)$	2
(c) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{\tan 4x}$	2

QUESTION 3 (9 Marks) – START A NEW PAGE**Marks**

- (a) Use Simpson's rule with five function values to estimate the volume of the solid formed by rotating the curve $y = \frac{2}{1+x^2}$ about the x -axis between 2 and -2 . 3
- (b) The flow rate of water first into and then out of a horse trough is R , where $R = 3\pi t(8 - t)$ litres/second.
- (i) When does water cease to flow into the horse trough and how much water has flowed into the trough in this time? Leave your answer in exact form. 3
- (ii) The trough was initially empty. Find how long the trough takes to empty again, and the rate at which the trough was then losing water. 3

QUESTION 4 (9 Marks) – START A NEW PAGE**Marks**

- (a) If (x_0, y_0) and (x_1, y_1) are two points on the line whose equation is $y = mx + k$, show that the distance between (x_0, y_0) and (x_1, y_1) in terms of x_0 and x_1 is $|x_0 - x_1|\sqrt{1 + m^2}$ units. 4
- (b) The area bounded by the curve $y = \frac{x}{\sqrt{1+x}}$, the line $x = 1$ and the x axis is rotated about the x axis. Find the volume of the generated solid of revolution. 5

QUESTION 5 (9 Marks) – START A NEW PAGE

Marks

(a) A function is defined by $f(x) = x - \frac{1}{x}$, $x > 0$.

(i) Show that $f(x)$ has an inverse and that this inverse is given by

$$g(x) = \frac{x}{2} + \sqrt{1 + \frac{x^2}{4}}.$$

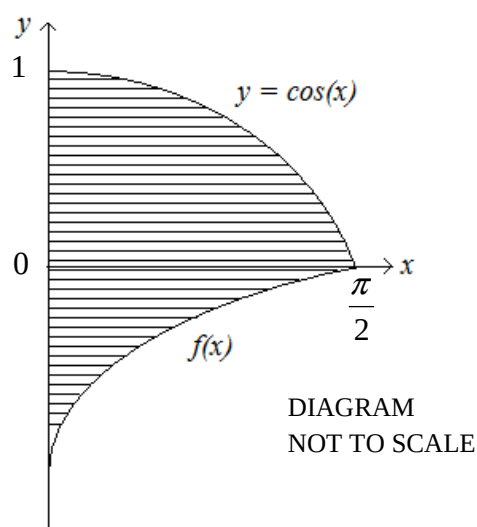
(ii) Sketch the curves $y = f(x)$ and $y = g(x)$ on the same set of axes.

State the domain and range of $g(x)$.

(b) The shaded area enclosed by the curves $y = \cos(x)$ and

$$y = f(x) = k \left(x - \frac{\pi}{2} \right)^2 \text{ and the } y\text{-axis is 4 square units.}$$

Find the exact value of k .



QUESTION 6 (9 Marks) – START A NEW PAGE

Marks

(a) (i) Differentiate $2x \tan^{-1}(2x) - \log_e \sqrt{1 + 4x^2}$ with respect to x .

(ii) Hence, evaluate $\int_0^{\frac{1}{2}} \tan^{-1}(2x) dx$ in exact terms.

(b) By finding the intercepts with the x and y -axes and any stationary points and by determining their nature, sketch the curve

$$y = e^x \cos x \text{ for } 0 \leq x \leq \frac{\pi}{2}, \text{ neatly.}$$

QUESTION 7 (9 Marks) – START A NEW PAGE**Marks**

(a) Write the general solution in radians for $\cos 2x = -\sin 3x$

3

(b) The Agriculture Faculty at James Ruse have a budget of \$ m to spend on constructing a rectangular lucerne paddock ABCD as shown in the diagram below. The side AB runs along the edge of the school dam and costs \$ n per metre to fence. The remaining three sides of the paddock cost \$ r per metre to fence.

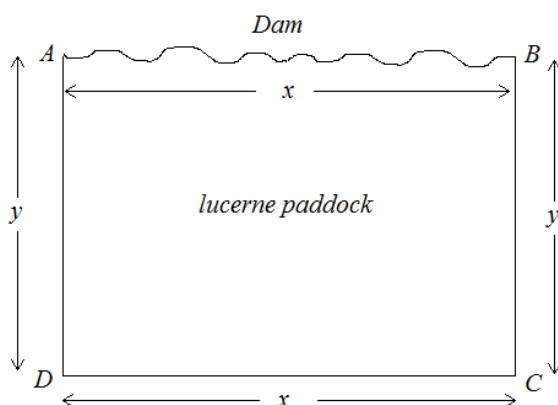


DIAGRAM
NOT TO SCALE

- (i) Find in terms of m , n , r and x , the width y of the paddock. **2**
- (ii) Hence, find the area of the paddock in terms of x , m , n and r . **1**
- (iii) Find the length (x) of the paddock in order to maximise the area of lucerne planted. **3**

END OF PAPER

TERM 4 ASSESSMENT 1, 2011
MATHEMATICS EX 1 : Question 1

Suggested Solutions	Marks	Marker's Comments
a) i) $2\tan^2 x - (1 + \tan^2 x) = 2$ $\tan^2 x = 3$ <u>$\tan x = \pm\sqrt{3}$</u> ii) $\tan x = \sqrt{3} \Rightarrow x = \pi/3 \text{ or } 4\pi/3$ $\tan x = -\sqrt{3} \Rightarrow x = 2\pi/3 \text{ or } 5\pi/3$ <u>$x = \pi/3, 2\pi/3, 4\pi/3 \text{ or } 5\pi/3$</u>	2	
b) $\frac{dy}{dx} = x^{1/2} - x^{-1/2}$ $y = \frac{2x^{3/2}}{3} - 2x^{1/2} + c$ When $x=4$, $y=3$ $\therefore 3 = \frac{16}{3} - 4 + c$ $c = 5/3$ $\therefore$ Eqn is <u>$y = \frac{2x^{3/2}}{3} - 2x^{1/2} + \frac{5}{3}$</u>	1	Many people did not read the question properly
c) Multiply by $(x+3)^2$ (>0) ($x \neq -3$) $x(x+3) > (x+3)^2$ $x^2 + 3x > x^2 + 6x + 9$ $-3x < -9$ <u>$x < -3$</u>	1	

MATHEMATICS Extension 1 : Question 2

Suggested Solutions

Marks

Marker's Comments

(a) $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$ Let $u^2 = x$ $u > 0$
 $\frac{dx}{du} = 2u$
 $dx = 2u du$

when $x=1$, $u=1$
 $x=4$, $u=2$
 $u > 0$

$\int_1^2 \frac{e^u \cdot 2u du}{u}$

$2 \int_1^2 e^u du$

$2[e^u]_1^2 = 2(e^2 - e)$

①

change
Limit values

①

Substitution

①

Integration and
answer.

(b) (i) $\sin^{-1} t$, $\cos^{-1} t$ $\sin^{-1}(1-t)$ are acute.

$-\frac{\pi}{2} < \sin^{-1} t < \frac{\pi}{2}$

$-1 < t < 1$

$0 < \cos^{-1} t < \frac{\pi}{2}$

$0 < t < 1$

$-\frac{\pi}{2} < \sin^{-1}(1-t) < \frac{\pi}{2}$

$0 < t < 2$

* Note.
acute angles
can be negative

Let $\alpha = \sin^{-1} t$ $\therefore \sin \alpha = t$ $-1 < t < 1$

$\beta = \cos^{-1} t$ $\therefore \cos \beta = t$ $0 < t < 1$

$\therefore 0 < t < 1$

$\cos^2 \alpha + \sin^2 \alpha = 1$ $\cos \alpha = +\sqrt{1-t^2}$
 $-\frac{\pi}{2} < \alpha < \frac{\pi}{2} \therefore \cos \alpha > 0$

$\cos^2 \beta + \sin^2 \beta = 1$ $\sin \beta = +\sqrt{1-t^2}$
 $0 < \beta < \frac{\pi}{2} \therefore \sin \beta > 0$

$\sin(\sin^{-1} t - \cos^{-1} t)$

$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$

$= t \times t - \sqrt{1-t^2} \times \sqrt{1-t^2}$

$= t^2 - 1 + t^2$

$= 2t^2 - 1$ $0 < t < 1$

($\frac{1}{2}$)

①

($\frac{1}{2}$)

Expansion
substitution
answer.

($-\frac{1}{2}$) if $0 < t < 1$
not mentioned
and part (ii)
not
completed.

MATHEMATICS Extension 1 : Question 2.....

Suggested Solutions

Marks

Marker's Comments

(b)(ii) $\sin^{-1} t - \cos^{-1} t = \sin^{-1}(1-t)$

consider

$$\sin(\sin^{-1} t - \cos^{-1} t) = \sin(\sin^{-1}(1-t))$$

$$2t^2 - 1 = n\pi + (-1)^n(1-t) \quad n \in \mathbb{Z}$$

But $0 < t < 1$ and $\sin^{-1}(1-t)$ is acute.

$$\therefore 2t^2 - 1 = 1 - t$$

$$2t^2 + t - 2 = 0$$

$$t = \frac{-1 \pm \sqrt{1 - 4(2)(-2)}}{4}$$

$$t = \frac{\sqrt{17}-1}{4} \quad \text{as } 0 < t < 1$$

only

①

quadratic equation

①/2

solution

①/2

one solution

①/2

for restriction $0 < t < 1$

(c) $\lim_{x \rightarrow 0} \frac{\sin 3x}{\tan 4x} = \frac{3}{4} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{4x}{\tan 4x}$

$$= \frac{3}{4} \times 1 \times 1$$

$$= \frac{3}{4}$$

①/2

must show "x's"

①/2

Must show "1's"

①

Answer

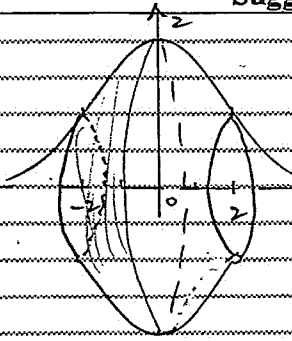
MATHEMATICS Extension 1 : Question 3

Suggested Solutions

Marks

Marker's Comments

3 (a)



$$\begin{aligned} \text{VOLUME} &= \pi \int_{-2}^2 y^2 dx \\ &= \pi \int_{-2}^2 \left(\frac{4}{1+x^2} \right)^2 dx \\ &= 2\pi \int_0^2 \frac{16}{(1+x^2)^2} dx \end{aligned}$$

(1/2)

(even function)

$$h = \frac{2 - (-2)}{4} = \frac{2-0}{2} = 1$$

$\frac{1}{2} + \left(\frac{1}{2}\right)$

x	-2	-1	0	1	2
y = $\frac{4}{(1+x^2)}$	$\frac{4}{5}$	1	4	1	$\frac{4}{5}$
	y ₁	y ₂	y ₃	y ₄	y ₅

$$\begin{aligned} \therefore V &= \pi \int_{-2}^2 y^2 dx = \pi \times \frac{h}{3} [y_1 + y_5 + 4(y_2 + y_4) + 2y_3] \\ &= \pi \times \frac{1}{3} \left[\frac{4}{5} + \frac{4}{5} + 4(1+1) + 2 \times 4 \right] \\ &= \frac{\pi}{3} \left[\frac{8}{5} + 8 + 8 \right] \end{aligned}$$

$$\therefore \text{VOLUME} = \frac{408\pi}{75} = \frac{136\pi}{25} = 5.44\pi \text{ units}^3$$

$$\text{Rate in} = 3\pi t(8-t)$$

* students are NOT reading the Q.

1 or equiv. formula

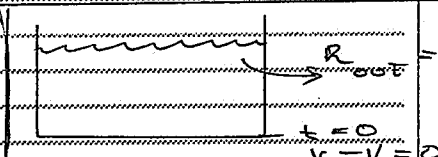
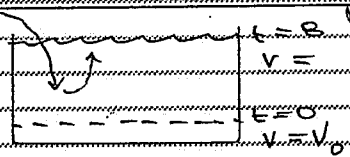
[3]

$-\frac{1}{2}$ if not

"volume = ... u"

$-\frac{1}{2}$ for 17.04...

(b) (i)



$$\text{cease flow when } R_{in} = \frac{dV}{dt} = 3\pi t(8-t) = 0$$

$$\therefore t = 0 \text{ or } t = 8$$

but $t \neq 0$: cease flow after 8 seconds

$$\begin{aligned} \text{VOLUME flowed in} &= \int_0^8 R dt = \int_0^8 3\pi(8t - t^2) dt \\ &= 3\pi \left[4t^2 - \frac{1}{3}t^3 \right]_0^8 \end{aligned}$$

$$= 3\pi \times \left(256 - \frac{512}{3} - 0 \right)$$

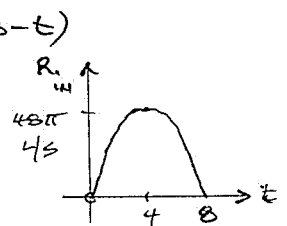
$$\therefore \text{VOLUME IN} = 256\pi L$$

$$\text{OR } V = \int 3\pi(8t - t^2) dt = 3\pi \left(4t^2 - \frac{1}{3}t^3 \right) + C$$

$$t = 0 \quad V = V_0 (\geq 0) \therefore C = V_0$$

$$\therefore V(t) = 24\pi t^2 - \pi t^3 + V_0$$

$$\therefore \text{VOLUME flowed in } V(8) - V(0) = 256\pi L.$$



1

[3]

MATHEMATICS Extension 1 : Question.....3

Suggested Solutions

Marks

Marker's Comments

Q 3(b)(ii) $R_{out} = -3\pi t(8-t)$

data: $t=0 \quad V=V_0=0$

$R = \frac{dV}{dt} = -3\pi t(8-t) = 3\pi t^2 - 24\pi t$

$\therefore V = \int 3\pi t^2 - 24\pi t \, dt =$

$V = \pi t^3 - 12\pi t^2 + C$

ie $V = \pi t^3 - 12\pi t^2 + V_0$

$\therefore V = \pi t^3 - 12\pi t^2$ ✓

Empty when $V=0$

$\therefore \pi t^2(t-12) = 0$

$\therefore t=0$ or $t=12$ but $t > 0$

$\therefore$ after 12 sec no volume in trough

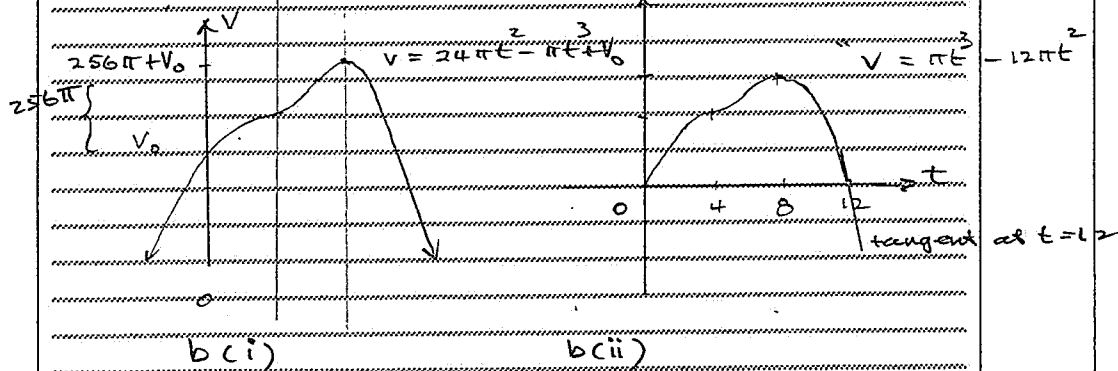
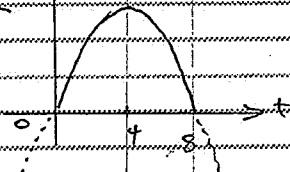
$\therefore$ Rate $= -3\pi \times 12(8-12) = 144\pi$

$\therefore$ flow rate out $= 144\pi \text{ L/s}$

• trough lossing at $144\pi \text{ L/s}$

• "Rate is $-144\pi \text{ L/s}$ "

Note $\frac{dV}{dt} =$



Relative sketch

$-\frac{1}{2}$ if wording in correct

3

MATHEMATICS Extension 1 : Question...4....

Suggested Solutions	Marks Awarded	Marker's Comments
a) Sub (x_0, y_0) into $y = mx + k$. $y_0 = mx_0 + k. \quad \text{--- eq 1}$ Sub (x_1, y_1) into equation $y_1 = mx_1 + k \quad \text{--- eq 2.}$ eq ① - ② $y_0 - y_1 = m(x_0 - x_1). \quad \text{--- eq 3 } \checkmark$ Now $d = \sqrt{(x_0 - x_1)^2 + (y_0 - y_1)^2} \quad \text{--- eq 4}$ Sub ③ into ④ $d = \sqrt{(x_0 - x_1)^2 + (y_0 - y_1)^2}$ $= \sqrt{(x_0 - x_1)^2 + m^2(x_0 - x_1)^2} \quad \checkmark$ $= \sqrt{(1+m^2)(x_0 - x_1)^2}$ as $\sqrt{(x_0 - x_1)^2} = x_0 - x_1$ reason compulsory $(\frac{1}{2} \text{ mark})$ $= \sqrt{(x_0 - x_1)^2} \cdot \sqrt{1+m^2}$ $= x_0 - x_1 \sqrt{1+m^2}$		
<u>Alternatively</u>, ② Some used $m = \frac{y_0 - y_1}{x_0 - x_1}$ $\therefore y_0 - y_1 = m(x_0 - x_1)$ ③ Some divided the distance formula. $d = \sqrt{(x_0 - x_1)^2 + (y_0 - y_1)^2}$ by $\sqrt{(x_0 - x_1)^2}$ and times by the same amount $d = \sqrt{\frac{(x_0 - x_1)^2}{(x_0 - x_1)^2} + \frac{y_0 - y_1}{(x_0 - x_1)^2} \times \sqrt{(x_0 - x_1)^2}}$		

4

MATHEMATICS Extension 1 : Question....

Suggested Solutions	Marks Awarded	Marker's Comments
b). $V = \pi \int_0^1 \frac{x^2}{1+x} dx$ ✓ $= \pi \int_0^1 \left((x-1) + \frac{1}{x+1} \right) dx$ ✓ $= \pi \left[\frac{x^2}{2} - x + \ln(x+1) \right]_0^1$ ✓ $= \pi \left[\frac{1}{2} - 1 + \ln 2 \right] - (0 - 0 + \ln 1)$ ✓ $= \pi \left[\ln 2 - \frac{1}{2} \right] \text{ unit}^3$ ✓ <u>Alternatively</u> Let $u = x+1$ <u>Method 2</u> , $x = u-1$ $x^2 = u^2 - 2u + 1$ when $x=0 \Rightarrow u=1$ $x=1 \Rightarrow u=2$ $\therefore V = \pi \int_1^2 \frac{u^2 - 2u + 1}{u} du$ <u>Method 3</u> , $V = \pi \int_0^1 \frac{x^2}{1+x} = \int_0^1 \frac{x^2 + 2x + 1}{x+1} - \frac{2x+1}{x+1} dx$ $= \int_0^1 (x+1) dx - \int_0^1 \frac{2x+2}{x+1} dx + \int_0^1 \frac{1}{x+1} dx$ OR . $V = \pi \int_0^1 \frac{x^2}{1+x} = \int_0^1 \frac{x^2-1+1}{1+x} dx = \int_0^1 \frac{x^2-1}{1+x} dx + \int_0^1 \frac{1}{1+x} dx$ Some used $\tan^{-1}$ <u>Note</u> instead of getting $\int \frac{x^2}{1+x} dx$ you have taken $\int \frac{x^2}{1+x^2} dx$ Here the maximum marks you got was 3 marks.		- show multiplied by π and squared - limits $\int_0^1$ * if no dx or limit lost $(-\frac{1}{2})$ <u>Long division</u> $\begin{array}{r} x-1 \\ x+1 \overline{) x^2+0x+0} \\ \underline{x^2+x} \\ 0-x-1 \\ \underline{-x-1} \\ 1 \end{array}$ $\therefore \frac{x^2}{x+1} = (x-1) + \frac{1}{x+1}$ <u>Some gave:</u> $-0.19314718\pi \text{ unit}^3$ or $0.606789763 \text{ unit}^3$ -accepted both. <u>Note</u>. * If no π and not squared. and simplified their answers working to get answer - got maximum of 2 marks * Marks taken off for simplifying their working making question easier.

Q5

(a) (i) $f(x) = x - \frac{1}{x}$ where $x > 0$

let $y = f(x)$

$\therefore y = x - \frac{1}{x}$

inverse function: $x = y - \frac{1}{y} \quad \therefore y > 0$

$yx = y^2 - 1$

$\therefore 0 = y^2 - yx - 1 \quad \frac{1}{2}$

$\therefore y = \frac{x \pm \sqrt{x^2 - 4(1)(-1)}}{2} \quad \frac{1}{2}$

Range is the $\frac{1}{2}$ domain of the original function

(3)

$y = \frac{x}{2} \pm \frac{\sqrt{x^2 + 4}}{2} \quad \frac{1}{2}$

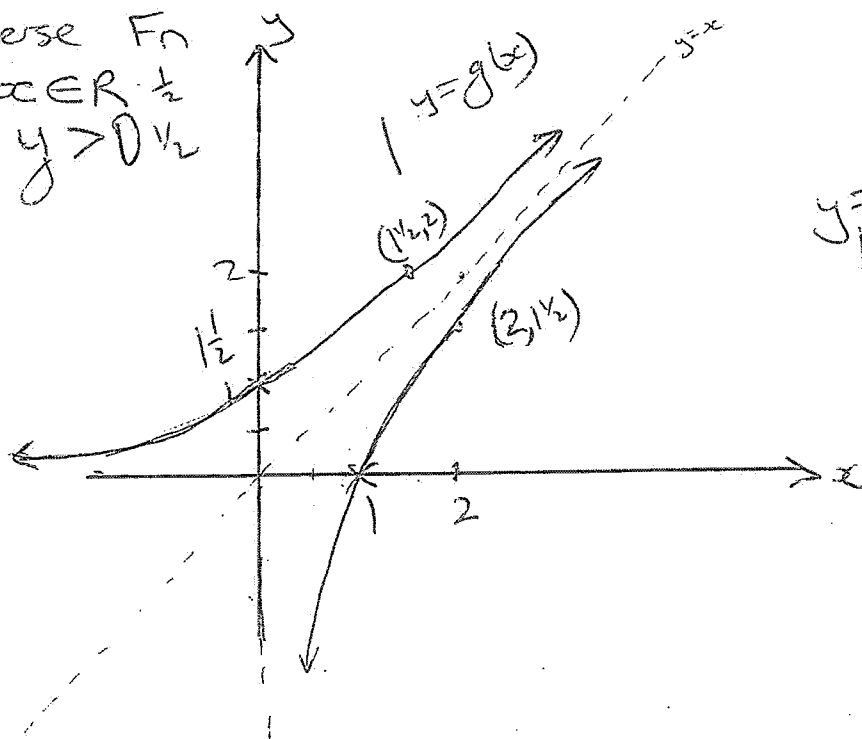
$\therefore y = \frac{x}{2} + \frac{\sqrt{x^2 + 4}}{2} \quad \text{as } y > 0$

$y = \frac{x}{2} + \frac{1}{2} \cdot 2\sqrt{\frac{x^2}{4} + 1} \quad \frac{1}{2}$

$\therefore g(x) = \frac{x}{2} + \sqrt{\frac{x^2}{4} + 1}$

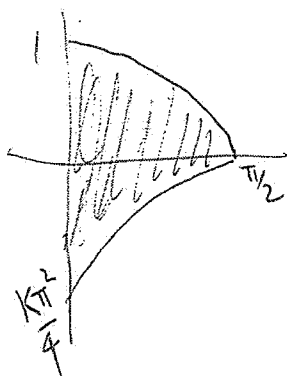
(ii)

Inverse Fn
D: $x \in \mathbb{R} \quad \frac{1}{2}$
R: $y > 0 \quad \frac{1}{2}$



(3)

(b)



$$\text{Area} = \int_0^{\pi/2} \cos x \, dx - K \int_0^{\pi/2} (x - \pi/2)^2 \, dx$$

$$4 = \left[\sin x \right]_0^{\pi/2} - \frac{K}{3} \left[(x - \pi/2)^3 \right]_0^{\pi/2}$$

$$4 = \sin \pi/2 - \sin 0 - \frac{K}{3} \left[\left(\frac{\pi}{2} - \pi/2 \right)^3 - (0 - \pi/2)^3 \right]$$

$$4 = 1 - \frac{K}{3} \left[\frac{\pi^3}{8} \right]$$

$$9 = -K \left(\frac{\pi^3}{8} \right)$$

$$\frac{-72}{\pi^3} = K$$

$$\therefore K = \frac{72}{\pi^3}$$

(3)

* $\frac{1}{2}$ off for each mistake.

Alternative method

$$\text{Area} = \int_0^{\pi/2} \cos x \, dx + \left| K \int_0^{\pi/2} (x - \pi/2)^2 \, dx \right|$$

very similar to above but when they remove the absolute value signs they need to take the negative case which nearly all students did not do!!

Yr 11 yrly 2011 Q6

4a a:) $y' = \frac{2x \cdot 2}{1+4x^2} + 2 \tan^{-1}(2x) - \frac{8x}{(1+4x^2)^{\frac{1}{2}}}$

If final answer is wrong -1 m

$y' = 2 \tan^{-1}(2x)$ #

ii) $\int_0^{\frac{1}{2}} \tan^{-1}(2x) dx = \frac{1}{2} \left[2x \tan^{-1} 2x - \ln \sqrt{1+4x^2} \right]_0^{\frac{1}{2}}$ 1 m MANY forgot $\frac{1}{2}$

$= \frac{1}{2} \tan^{-1} 1 - \frac{1}{4} \ln 2 + \frac{\ln 1}{4}$

$= \frac{\pi}{8} - \frac{1}{4} \ln 2$ 1 m #

5m b) $y = e^x \cos x \quad 0 \leq x \leq \frac{\pi}{2}$

$y' = e^x (\cos x - \sin x)$ $\frac{1}{2}$ m

SP $y' = 0$ when $\cos x = \sin x$ $\frac{1}{2}$ m

$\therefore \tan x = 1 \quad x = \frac{\pi}{4}, \quad y = \frac{e^{\frac{\pi}{4}}}{\sqrt{2}} (\doteq 1.551)$ $\frac{1}{2}$ m

$y'' = e^x [-\sin x - \cos x] + e^x (\cos x - \sin x)$

$y'' = -2e^x \sin x$ $\frac{1}{2}$ m

when $x = \frac{\pi}{4} \quad y'' = -\frac{2e^{\frac{\pi}{4}}}{\sqrt{2}} < 0 \quad \therefore$ concave down

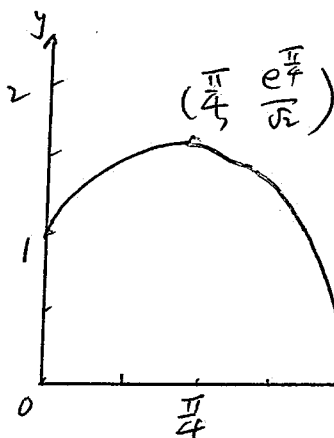
$\frac{1}{2}$ m rel max at $(\frac{\pi}{4}, \frac{e^{\frac{\pi}{4}}}{\sqrt{2}})$

when $x=0, y=1$ $\frac{1}{2}$ m

$y=0, x=\frac{\pi}{2}$ $\frac{1}{2}$ m

MAX 4 out of 5 for part b

for $y = e \cos \frac{\pi}{4}$ without simplify to $e^{\frac{\pi}{4}}/\sqrt{2}$



many have wrong scale $\frac{1}{2}$ m or forgot origin $\frac{1}{2}$ m

1 m for shape with labelled T.P. intercepts $\frac{1}{2}$ m scale with correct graph 2 m

Suggested Solutions	Marks	Marker's Comments
Q7(a) $\cos 2x = -\sin 3x$ $= \sin(-3x)$ odd function $= \cos\left[\frac{\pi}{2} - (-3x)\right]$ $\cos 2x = \cos\left(\frac{\pi}{2} + 3x\right)$ $\therefore 2x = 2m\pi \pm \left(\frac{\pi}{2} + 3x\right)$ $2x = 2m\pi + \frac{\pi}{2} + 3x$ or $2x = 2m\pi - \frac{\pi}{2} - 3x$ $\therefore x = \begin{cases} -(4m+1)\frac{\pi}{10} \\ (4m-1)\frac{\pi}{10} \end{cases}, m \in \mathbb{Z}$ • IF $\sin\left(\frac{\pi}{2} - 2x\right) = \sin(-3x)$ from $\cos 2x = -\sin 3x$ $x = \frac{(2m-1)\pi}{2[3(-1)^m - 2]}$ or $\frac{[2m + (-1)^m]\pi}{2[2(-1)^m - 3]}$ • IF $-\cos 2x = \sin 3x$ $= \cos\left(\frac{\pi}{2} - 3x\right)$ $\therefore x = \begin{cases} (4m-1)\frac{\pi}{10} \\ (3-4m)\frac{\pi}{10} \end{cases}, m \in \mathbb{Z}$	1 1 each	3
(b) (i) x at $\\$m$ $m = nr + (n+r)x + 2ry$ $m = (n+r)x + 2ry$ $\therefore y = \frac{m - (n+r)x}{2r}$	1 1	2
(ii) $A = xy$ — (1) $y = \frac{m - (n+r)x}{2r}$ — (2) $\therefore A(x) = x \left[\frac{m - (n+r)x}{2r} \right]$ $A(x) = \frac{m}{2r}x - \frac{(n+r)}{2r}x^2$	1	11

(iii) P.T.O.

Suggested Solutions

Marks

Marker's Comments

Q 7(b) (iii)

$$A(x) = \frac{m}{2r}x - \frac{(n+r)}{2r}x^2$$

METHOD 1 From Quadratic Polynomials

$$As \ a = -\frac{(n+r)}{2r} < 0$$

$\therefore$ concave downwards

Hence a maximum value will exist

$$\text{at } x = -\frac{b}{2a} \quad \text{or} \quad \frac{\alpha + \beta}{2}$$

$$\text{i.e. } x = \frac{m}{2(n+r)}$$

$$\therefore y = \frac{m}{2r} - \frac{n+r}{2r} \times \frac{m}{2(n+r)} = \frac{m}{4r}$$

$$A = \frac{m}{2(n+r)} \times \frac{m}{4r} = \frac{m^2}{8r(n+r)}$$

$\therefore$ abs max Area when $x = \frac{m}{2(n+r)}$

METHOD 2 $A = \frac{m}{2r}x - \frac{(n+r)}{2r}x^2$

$$\frac{dA}{dx} = \frac{m}{2r} - \frac{(n+r)}{r}x$$

$$\frac{d^2A}{dx^2} = -\frac{(n+r)}{r}$$

For possible max/min values of A to occur $\frac{dA}{dx} = 0$

$$\therefore \frac{m}{2r} - \frac{(n+r)}{r}x = 0$$

$$\therefore x = \frac{m}{2(n+r)}$$

$$y = \frac{m}{4r}, \quad A = \frac{m^2}{8r(n+r)}$$

TEST nature

$$As \ \frac{d^2A}{dx^2} = -\frac{(n+r)}{r} < 0 \quad ; \ r \text{ and } n > 0$$

$\therefore$ concave downwards and since a SP at $x = \frac{m}{2(n+r)}$

$\therefore$ a Rel. max TP at $x = \frac{m}{2(n+r)}$

Since continuous and no other SPs for $x \geq 0$

$\therefore$ the abs. max. Area occurs when $x = \frac{m}{2(n+r)}$

OR

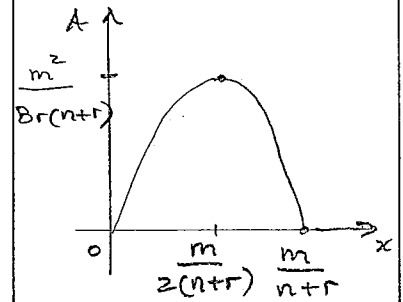
$$x \quad \frac{m}{4(n+r)} \quad \frac{m}{2(n+r)} \quad \frac{m}{(n+r)}$$

$$\begin{array}{ccccccc} \frac{dA}{dx} & \frac{m}{4r} & 0 & -\frac{m}{2r} \\ & \oplus & & \ominus \\ & \nearrow & & \searrow \end{array}$$

$\therefore$ a Rel. max TP at $x = \frac{m}{2(n+r)}$

etc. see above

$$0 \leq x \leq \frac{m}{n+r}$$



$$a = -\frac{(n+r)}{2r}; \quad b = \frac{m}{2r}$$

$$\alpha = 0; \quad \beta = \frac{m}{n+r}$$

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